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Engineering Biodegradable Sodium Alginate Films with Dragon Fruit Peel Powder and MgO Nanoparticles

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ABSTRACT

This study investigated the effect of the addition of dragon fruit peel powder (DFP) and MgO nanoparticles on sodium alginate (SA) films. The control film (S1 film), DFP-enriched (S2 film), MgO nanoparticles-enhanced (S3 film), and a film combining both MgO nanoparticles and DFP (S4 film) were evaluated. The tensile strength increased by 252.43, 290.27, and 645.13% for the S2, S3 and S4 films, respectively. These enhancements were accompanied by a marked improvement in the UV-blocking property, thermal stability, antioxidant activity,2,2′-azino-bis-3-ethylbenzothiazoline-6-sulfonic acid (ABTS) radical scavenging activity, opacity and thickness. The water vapor permeability and transparency were decreased to 3.45×10^{-10} g^{-1} Pa^{-1} S^{-1} and 1.82% for the S4 film. An environmental scanning electron microscope (ESEM) study revealed a notable increase in surface roughness with the incorporation of DFP. X-ray diffraction (XRD) analysis indicated that the MgO nanoparticles possessed a crystallographic structure that was compatible with the SA film matrix, with the S4 films exhibiting the highest crystallinity index (28.92%). The overall migration study showed that all SA films complied with the overall migration limit of 60 mg/kg of simulant for fatty food simulants. The biodegradability test revealed complete degradation of the S1, S3, and S4 films before 20 days while the S2 film exhibited a substantial mass loss of 84.99% within the same 20-day timeframe.

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anti-oxidant activity; biodegradable packaging film; dragon fruit peel powder; MgO nanoparticles; UV-blocking property; tensile strength improvement

1. Introduction

In the last 50 years the production of plastic has increased by 20 times, and scientists have predicted that plastic production may surpass 500 million tons/year in the year 2050.[1] After utilization of the plastic more than 60% is thrown into landfill sites. After getting broken down into microplastic, much of it enters into the marine ecosystem.[2] When consumed by marine animals and later by humans, the microplastic can be the sole reason for physiochemical damage.[3] Microplastic can enter the human body through ingestion, inhalation and dermal routes and can cause oxidative stress, cytotoxicity, inflammation, cancer, disruption of immune function, neurotoxicity, and increased risk of neurodegenerative diseases.[4] High density polyethylene (HDPE) and low density polyethylene (LDPE) are used for rigid and flexible packaging purpose, respectively.[5] 40% of all the polyethylene (PE) produced is utilized for the purposes of packaging.[6] Therefore, a biodegradable and nontoxic alternative to PE is needed for packaging purposes.

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Biodegradable plastics such as, polycaprolactone, polybutylene succinate, polylactide aliphatic copolymer, polyglycolide, polyvinyl alcohol, and polypropylene carbonate, etc. are used for packaging purposes, but their production more or less relies on fossil fuel, which contributes to greenhouse emission. The advantage of using bioactive packaging (over the above mentioned plastics) is that they can enhance the shelf life of fruits and vegetables.[7]

Ever-growing public concern for the environment has enabled the recent research drive in the field of production of edible packaging from biopolymers. Of all the biopolymers available, polysodium alginate (PSA) is of great interest because of its nontoxic and water-soluble nature. Sodium alginate (SA) is a linear polysaccharide mainly obtained from brown seaweed. It is made up of two hexuronic acid residues: (1) β-D mannuronate and (2) α-L-guluronate, which are linked together by 1,4-glycosidic bonds.[8]

Previous literature in the field of the bioactive packaging has mainly focused on the addition of various kinds of waste peel and leaf extracts, such as *Azollapinnata* leaves extract,[9] *Syzygiumcumini* seeds extract,[10] and guava leaf extract,[11] etc, into SA based biofilms to enhance their properties. However, a recent research trend focuses on the possibility of directly incorporating the waste peel and leaf powder into the biopolymer matrix. It is well known that waste peel powder, such as orange peel powder,[12] pomegranate peel powder,[13] banana peel powder [14] contains cellulose, hemicellulose, dietary fibers, protein, carbohydrates, and bioactive compounds. Recent studies have been carried out with the central aim of the addition of various peel powders (like the above mentioned) into different kinds of biopolymers, like chitosan/polyvinyl alcohol (PVA),[12] gelatin [13] and chitosan.[14] For instance, orange peel powder addition to the chitosan/PVA film showed an improvement in its antioxidant activity, UV-light blocking properties, and a decrease in oxygen permeability.[12]

However, we have found only rare research articles in which any type of peel powder was added to fabricate SA based biofilms. Yun et al.[15] utilized different types of citrus peel powder and sodium alginate (as an adhesive agent) to fabricate biofilm. The addition of the peel powder improved the antioxidant and antibacterial properties. Mandarin peel powder based film had the highest oxygen and light barrier properties, and better suitability for enhancing corn oil shelf life. Apart from this, to the best of our knowledge, there has been no other study in which any kind of peel powder was utilized to fabricate SA films. Therefore, we choose dragon fruit peel powder (DFP) to fabricate SA biofilms. The reason was that the DFP is known to contain dietary fibers, carotenoids, phenolic compounds, betacyanin compounds, betaxanthin compounds, vitamin C, protein, etc.[16] Our hypothesis was that these components in DFP could help improve the properties of the SA films, the reason being the similar components but in different combinations in citrus peel powder improved SA film properties.[15]

As per previous literature, various nanomaterials, like SiO_2,[17] ZnO,[18] CuO,[19] silver,[20] and TiO_2,[21] etc, have been added to SA biofilms. The effect of the addition of a combination of propolis extract and SiO_2 to SA film has also been reported.[17] A 51.9% rise in tensile strength was observed without any significant decrease in elongation at break. MgO is colorless, and can be manufactured on a large scale at low cost. The U. S. Food and Drug Administration has recognized the safe use of MgO nanoparticles in food applications.[22] However, we found no literature in which MgO nanomaterial was utilized to fabricate SA biofilms. Therefore, MgO nanoparticles were chosen as an ideal candidate for the current study.

The aim of the current study was to fabricate and characterize SA biofilms containing a combination of dragon fruit peel powder and MgO nanoparticles. It was our hypothesis that the addition of DFP and MgO nanoparticles to the SA would result in biofilms with enhanced mechanical, antioxidant, antimicrobial, and water absorption properties. Initially, a screening study was carried out to find the films with the optimum concentration of MgO nanoparticles and peel powder. The characterization of the DFP was also carried out. Then, the thermal properties (TGA), FTIR, tensile, elongation at strength, XRD, ESEM, and contact angle analyses of the peel powder-nano MgO SA (PPMSA) biofilms were carried out. In addition to this, biodegradability tests of all of the biofilms were also done.

2. Materials and methods

2.1. Materials

SA was purchased from Loba Chemie Pvt. Ltd., India. Red dragon fruits were obtained from a local market in Surat, India. Glycerol was purchased from Finar Ltd., India. Magnesium oxide nanoparticles (average particle size of 50 nm), and 2,2′-azino-bis-(3-ethylbenzothiazoline-6-sulfonic acid) (ABTS) were obtained from Sigma Aldrich Ltd., India.

2.2. Preparation, proximate analysis, and extraction of DFP

The red dragon fruit were washed with tap water and peeled using a steel knife. The red dragon fruit peels were cut into small pieces (3×2 cm), kept in a deep freezer overnight, freeze-dried for 16 h, ground, and passed through a BSS 150 mesh size sieve to obtain red dragon fruit peel powder (DFP). The DFP was kept in a polyethylene bag at $-4\,^{\circ}$C in a dark room till further use. As per the established procedure in the literature, the moisture, ash, crude fiber, protein and crude fat content were measured.[23] The DFP powder was subjected to extraction using water as a solvent using a previously described method.[24] Then, the measurement of pectin content, and total phenolic, total betacyanin and total betaxanthin content of the extract was carried out.

2.3. SA based biofilm preparation method

SA based biofilm was fabricated using a previously described method with modification.[15] 2.5 g SA was added to 150 mL deionized water and kept stirring overnight. Then 0.75 mL glycerol, 0.24 g peel powder and 0.12 g MgO nanoparticles were added and stirred for 30 min. The solution was sonicated for 15 min. Then 25 mL of the suspension was cast onto a 25 cm diameter acrylic dish and kept overnight under a fan at room temperature. The films were then peeled off and kept in a desiccator at 50% humidity at room conditions for 2 days. It was observed that all of the SA biofilms were easier to peel off an acrylic disk than a glass petri dish.

2.4. Pectin, phenolic, betacyanin, and betaxanthin contents of the DFP

The pectin content of the DFP was determined by modifying a previously described method.[25] 1 g DFP was added to 30 mL deionized water and heated to 85 °C for 32 min.

Citric acid was utilized to maintain the pH in the range of 3 to 4. Then it was filtered using Whatman filter paper no. 1, and the supernatant was subjected to centrifugation. The clear supernatant was mixed with 95% ethanol and then with 50% ethanol to precipitate the pectin. The pectin was filtered from the solution and then it was dried in an oven at 60 °C overnight. The yield of pectin was calculated using the following equation:

$$Y_{pectin} = \left(\frac{P_{ext}}{D_i}\right) \times 100 \quad (1)$$

where Y_{pectin} is the yield of pectin (%), P_{ext} is the amount of dried pectin obtained (g) and D_i is the initial weight of the DFP (g).

The betacyanin and betaxanthin contents were measured by the spectrophotometric method.[26] The phenolics content (PC) was measured using the Folin-Ciocalteu (FC) method.[27] Betanin and iso-betanin pigment compounds were quantified by HPLC analysis using our previously described method.[24]

2.5. Mechanical analysis of the films

The tensile strength (TS) and % elongation at break of the SA film specimens (100 × 20 mm) were quantified through the utilization of a universal testing apparatus (Instron-5967, Instron Corp., USA). The initial gauge length employed was set at 60 mm. The test was performed at a crosshead speed of 5 mm per min. To determine the thickness of the SA films a portable thickness gauge meter with a precision margin of 0.01 mm was used. The average thickness was derived from five randomly selected measurements conducted on each of three film samples.

2.6. FTIR

The SA film's IR spectra were recorded on an IR spectrometer (Alpha II FTIR, Bruker Corp., Germany) with a platinum ATR single reflection diamond sampling module. The sample was directly put into the sampling module. The recorded IR spectra of the SA films were obtained by averaging data points from 16 scans per sample within the wavenumber range of 600–4000 cm^{-1} and at a resolution of 4 cm^{-1}. The Optics User Software (OPUS Version 6.5, Bruker Corp., Germany) was used for analysis.

2.7. Thermal analysis of the films

A thermal gravimetric analyzer (TGA-7 Pyris 1, Perkin Elmer Inc., USA) was used to investigate the thermal properties of the SA films. 50 mg samples were subjected to heating within a temperature range spanning from 30 °C to 600 °C with a ramp-up rate of 10 °C/min, all while maintaining a nitrogen environment.

2.8. ESEM analysis

Surface morphology observations were carried out using an ESEM (Quanta 200, Field Electron and Ion Co., Netherlands) instrument. To enhance the sample's conductivity, a

thin layer of gold (Au) coating was applied, involving a cycle time of 10 s and achieving a coating thickness in the range of 2–5 nanometers.

2.9. XRD analysis

To assess the crystal structure of the SA films and MgO nanoparticles, XRD analysis was conducted employing a Miniflex instrument, Rigaku Corp., Japan with Cu K_α radiation (1.54065 Å). The XRD spectra were acquired over the 4 to 70° (2θ) range at a scanning rate of 5°/minute. The crystallinity index (CI%) was measured using the following equation reported previously:[28]

$$\mathrm{CI}(\%) = \left(\frac{\mathrm{A}_1}{\mathrm{A}_1 + \mathrm{A}_2}\right) \times 100 \tag{2}$$

where A_1 and A_2 are the areas of all crystalline peaks and the amorphous region, respectively.

2.10. UV-spectra analysis, film opacity and transparency of the films

The opacity and transparency were quantified at 500 and 600 nm, respectively, using previously described method.[12] A microplate reader (SPECTROstar nano, BMG Labtech Ltd., Germany) was employed to record the transmittance spectra of the SA films. Prior to this the SA films were segmented into pieces measuring 10 × 40 mm, and these film pieces were directly inserted into the test cell of the microplate reader. Subsequently, the spectra were registered over the wavelength range of 200–800 nm, while an empty cell was used as a blank reference.

The film opacity was measured using the following equation:

$$\text{Film opacity} = \frac{\mathrm{Abs}_{500}}{\mathrm{x}} \tag{3}$$

where Abs_{500} and x are the absorbance values at 500 nm and film thickness (mm), respectively.

The transparency was measured using the following equation:

$$\text{Trasnparency} = \frac{\log(\%\mathrm{T}_{600})}{\mathrm{x}} \tag{4}$$

where $\%T_{600}$ is the % transmittance at 600 nm and x is the film thickness (mm).

2.11. Moisture content (MC)

The MC (%) was measured using a previously described method.[9] The samples were subjected to drying in an air oven at a fixed temperature of 110 °C for a time duration of 24 h, and the MC (%) of the films was calculated using the following equation:

$$MC(\%) = \frac{W_1 - W_2}{W_1} \times 100 \tag{5}$$

where W_1 (initial weight) and W_2 (final weight) pertain to the weights of the films at the beginning and end of the measurement, respectively.

2.12. Water vapor permeability (WVP) of the film

The WVP was also measured using a method described previously.[29] A fixed quantity of dried calcium chloride ($CaCl_2$) was placed within a glass vial. Subsequently, a film specimen was hermetically sealed over the top of the glass vials. These sealed glass vials were then placed within the confines of a desiccator, which was configured to accommodate a 50 mL beaker at the base containing a saturated sodium chloride (NaCl) solution. The mass of the glass vials was recorded until a constant weight was attained. The WVP was ascertained employing the following equation:

$$WVP = \left(\frac{\Delta w \times D}{A \times t \times \Delta P}\right) \tag{6}$$

where Δw is the change in weight (g), D is the film thickness (m), A is the area of film (m^2), t signifies the time duration (s), and ΔP is the difference in water vapor pressure (Pa) taken as 3169 Pa at 25 °C.[30] The ΔP value is based on the assumption that the headspace at top is completely saturated with water vapor, and a completely dry environment at the bottom of the glass vial is provided by the dried $CaCl_2$.

2.13. Contact angle measurement and wet stability study of the SA films

The contact angle of the water on all of the biofilms was measured using a contact angle meter (Model: OCA 15EC, Data Physics Corp., USA), and static and dynamic contact angle measurement (SCA-20, Data Physics Corp., USA) software. The dynamic contact angle study of the SA films was performed for 1 min, in which the contact angle was measured every 10 s to measure the wettability of the film. A micro-needle was used to obtain a water droplet of 1 µL, which was poured on the top surface of the SA films. Then, the contact angle was recorded using the SCA-20 software. To investigate the stability of the biofilms in the wet environment, the SA films were cut into square 50 × 50 mm pieces, and were subjected to immersion in 40 mL water for 20 min. Then the morphology of the films was observed by photographic evidence.

2.14. Antioxidant activity of the films

The assessment of the antioxidant activity of the SA films followed a method reported previously in the literature with a few modifications.[9] A 30 mg film sample was dissolved in 3 mL distilled water and centrifuged at 6000 rpm for 10 min to obtain the supernatant extract. The ABTS stock solution was prepared by mixing 7 mM/L ABTS with 145 mM/L potassium persulfate solution. The solution was then kept in the dark room for 12 h. Next, 1900 μLof the ABTS stock solutionand 100 μL supernatant were mixed and kept in a dark room for 30 min at 28 °C. The absorbance was then measured using a microplate reader (SPECTROstar nano, BMG Labtech Ltd., Germany) at 732 nm. The ABTS radical scavenging activity were measured using the following equation:

$$\text{ABTS radical scavenging activity } (\%) = \left(1 - \frac{A_1}{A_0}\right) \times 100 \tag{7}$$

where A_1 and A_0 are the absorbance of the sample and blank, respectively.

2.15. Overall migration study

The overall migration (OM) study of the SA film was carried out by utilizing two distinct food simulants: 50% ethanol, designated as 'simulant D1' (representative of alcoholic food beverages), and isooctane, designated as 'simulant D2' (simulating a fatty food environment). The procedure involved a precise weighing of film strips using an analytical weighing balance (ME54/A04, Mettler Toledo India Pvt. Ltd., India, accuracy ±0.0001 g), followed by their immersion in a glass vial containing the respective simulants. Consistent with EN-1186-1 guidelines, the film samples underwent an incubation period of 10 days in 50% ethanol (at a controlled temperature of 40 °C) and 2 days in isooctane (maintained at 20 °C). Then the films were carefully removed, the simulant was evaporated in a heating oven, and the residue was weighed.[31] The test was performed in triplicates, and average readings were reported.

2.16. Biodegradation study of the films

The biodegradation test was performed to evaluate the ability of the films to undergo natural degradation when exposed to organic compost. The organic compost was filled up to 15 cm height in a container. The films were cut into $2.5 \times 2.5\ cm^2$ pieces and kept at a 7.5 cm height in the container. Water was sprinkled twice a day to maintain a 40%–50% moisture content of the organic compost. At every 5-day interval the films were removed and dust was removed from them. The weight loss was noted, and % weight loss over this time interval was calculated.[32] The films were also photographed at these time intervals for observation of their physical appearance.

2.17. Statistical analysis

Statistical analysis of the various results was conducted through the utilization of Minitab software (Version 18, Minitab LLC, USA). Subsequently, the data underwent examination *via* the one-way analysis of variance (ANOVA) and were subjected to Tukey's post-hoc analysis. It is essential to note that within the tabulated results (Tables 1 and 2) the presence of distinct alphabetical characters within a shared column signifies statistically significant distinctions with a significance level below 0.05. The presented figures denote the mean ± standard deviation and were derived from 3 samples.

3. Results and discussion

3.1. A proximate analysis of the DFP powder and its bioactive compounds

The crude fiber, protein, crude fat, moisture and ash content of the DFP were found to be 17.02%, 9.12%, 1.66%, 0.63%, and 9.88%, respectively. The protein, crude fiber, and pectin collectively constitute the structural framework of the PPMSA films, whereas the crude fat and polyphenols play distinct roles in enhancing the hydrophobicity and bioactive properties of the films.[33] The crude fiber method typically measures cellulose, hemicellulose and lignin content. Although it underestimates the actual content of fibers, the values of crude fiber obtained can give a rough idea about the amount of the cellulose,

Table 1. Film thickness, TS, elongation at break (%), MC (%), film opacity (AU nm/mm), transparency (%), WVP ($\times 10^{-10}$) $g^{-1} Pa^{-1} S^{-1}$, and ABTS radical scavenging activity (%) of the S1, S2, S3, and S4 films.

Sam-ple	Thickness (mm)	TS (MPa)	Elongation at break (%)	MC (%)	Film opacity (AU nm/mm)	Transparency (%)	WVP ($\times 10^{-10}$) $g^{-1} Pa^{-1} S^{-1}$	ABTS radical scavenging activity (%)
S1 film	0.0540 ± 0.0026^{a}	2.88 ± 0.05^{a}	105.78 ± 0.11^{a}	20.93 ± 0.77^{b}	13.58 ± 0.54^{a}	26.29 ± 1.31^{a}	3.55 ± 0.04^{ab}	22.53 ± 1.47^{a}
S2 film	0.0686 ± 0.0015^{a}	10.15 ± 0.42^{b}	106.09 ± 0.24^{a}	19.35 ± 0.77^{b}	16.61 ± 0.54^{b}	16.18 ± 0.03^{b}	3.28 ± 0.08^{b}	39.12 ± 1.95^{b}
S3 film	0.1033 ± 0.01387^{b}	11.24 ± 0.75^{b}	106.73 ± 0.44^{a}	24.14 ± 0.81^{a}	16.14 ± 0.53^{b}	3.85 ± 0.05^{c}	3.63 ± 0.08^{a}	73.48 ± 0.52^{c}
S4 film	0.1066 ± 0.0083^{b}	21.46 ± 1.91^{c}	104.46 ± 2.71^{a}	21.00 ± 0.19^{b}	20.14 ± 0.61^{c}	1.82 ± 0.05^{d}	3.45 ± 0.04^{ab}	96.16 ± 1.57^{d}

Table 2. OM values of the various SA films for 50% ethanol and isooctane at room temperature.

Sample	OM for 50% ethanol (mg/kg of simulant)	OM for isooctane (mg/kg of simulant)
S1 film	3399.26 ± 67.14^{a}	n.d
S2 film	4703.29 ± 495.35^{b}	n.d
S3 film	6593.40 ± 288.23^{c}	n.d
S4 film	7487.17 ± 285.99^{d}	n.d

hemicellulose, and lignin content.[34] The peel part of the dragon fruit is considered inedible, and proximate analysis reports in the literature vary greatly. Biswas et al.[16] reported the moisture, protein, fat, and ash content of DFP to be 9.99%, 6.03%, 6.14%, and 4.34%, respectively. The authors also reported the dietary fiber content to be 59.83 g/100 g of peel powder. Madane et al.[35] reported the lipid, ash, and protein content of DFP to be 4.48%, 2.34%, and 10.36%, respectively. Dragon fruit peel is known to be rich in bioactive compounds. In our study, pectin and total phenolic content were found to be 5.85% and 2.82 mg gallic acid equivalent/g, while total betacyanin and total betaxanthin content were found to be 8.12 mg/g and 3.15 mg/g, respectively, of the DFP powder. Using HPLC analysis, two pigments were identified, betanin and iso-betanin, and were quantified to be 8.73/10 g dry weight and 0.22 mg/10 g dry weight, respectively. In the literature, different values for the pectin content in dragon fruit peel have been reported, such as 9.38% and 7.5%.[25,36] Total betacyanin contents of 9.3 mg/10 g, 8.4 mg/10 g, and 13.1 mg/10 g for Mexican, Indonesian, and Spanish species of DFP have been reported.[37] Li et al.[38] reported the betacyanin and betaxanthin content of red dragon fruit peel to be 29.43 and 20.60 mg/L, respectively. The reason for the differences in values of the proximate analysis of the DFP and its bioactive compounds of the current study with the literature could be variability of fruit, ripeness of fruit at the time of harvesting, geographical location, variations in drying method, grinding, sample homogenizing, and storage conditions, methodological differences, and extraction efficacy.[39]

3.2. Screening study of SA based biofilms

As shown in Figure 1, the initial screening study was conducted to ensure the quantity of the optimum amount of DFP and MgO nanoparticles to be added. The tensile strength (TS) was kept as a single parameter for choosing the optimum amount of DFP and MgO nanoparticles. Films with higher TS were chosen for the next study. In the recent literature, a similar screening study based on the TS as the response parameter was also conducted. The authors found that the TS was dependent on the concentration of MgO nanoparticles and tea polyphenols added to fabricate a potato starch film, and the film with the highest TS was considered as a film with the optimum concentration of MgO nanoparticles and tea polyphenols.[40] Figure 1(a) illustrates the effect of the variation in the concentration of MgO nanoparticles from 2.5% to 20% in SA on the TS, where DFP was not added. Figure 1(b) demonstrates the effect of the variation in concentration of DFP from 5% to 40% in SA on the TS, without addition of MgO nanoparticles. With an increasing concentration of DFP from 5% to 20%, the TS increased from 5.34 ± 0.3 MPa to 10.15 ± 0.42 MPa, while further increasing the concentration of DFP resulted in a decrease in TS. This may be because a low concentration of DFP may get easily dispersed to form a tighter polymer network, whereas a higher concentration may form aggregates, which

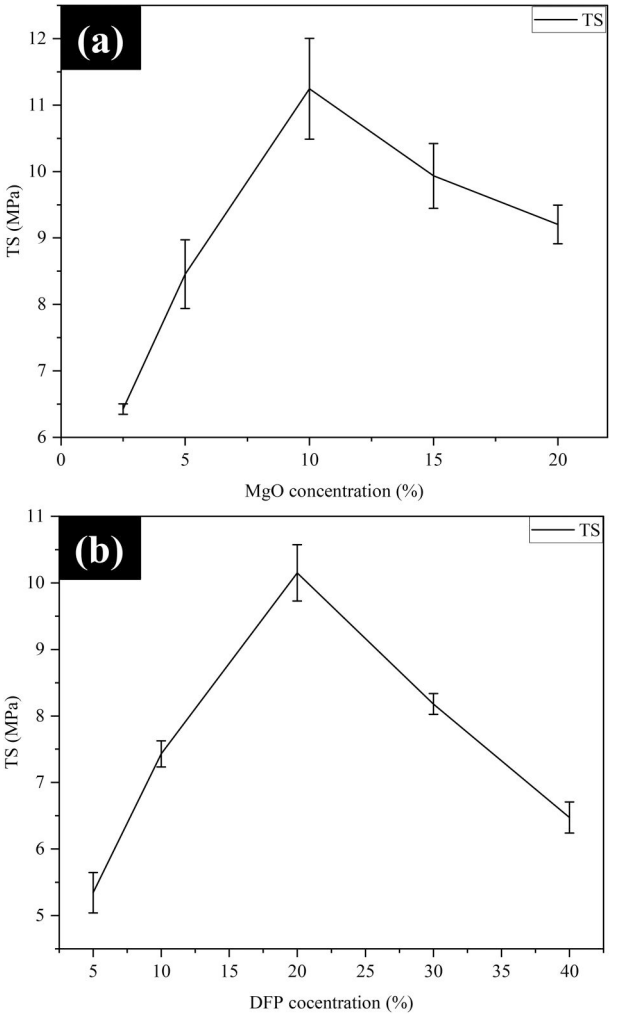


Figure 1. Effect of change in MgO nanoparticles (a) and DFP concentration (b) on TS.

may disrupt the initially formed polymer structure.[41] Increasing the MgO nanoparticles concentration from 2.5% to 10% increased the TS up to 11.24 ± 0.75 MPa, followed by a decrease in TS on further increasing the MgO nanoparticles concentration. The lower MgO nanoparticles concentration may reinforce the polymer network resulting in increasing TS, while further increasing the concentration could disrupt the compact polymer network. At the optimum concentration of 10% MgO and 20% DFP, the TS value increased significantly, up to 21.46 ± 1.91 MPa. The addition of MgO and DFP thus brought a synergistic effect. Similar results of the optimum concentration have been reported in the literature as well, in which a 24 MPa TS was obtained when 0.6% tea polyphenols and 1.5% MgO nanoparticles were added simultaneously to fabricate a potato starch-based biofilm,[40] although the amounts added were considerably different than in our case. In addition, the interfacial adhesion of polyphenols and nanoparticles was found to be responsible for an increase in TS of chitosan film by the addition of ZnO nanoparticles and mulberry polyphenols.[42] Table 3 shows the composition of the various SA films,

Table 3. Sample code and corresponding composition.

Sample code	SA (g)	Glycerol (mL)	Water (mL)	MgO nanoparticles (g) (% wt of SA)	DFP (g) (% wt of SA)
S1 film	1.2	0.36	60	0	0
S2 film	1.2	0.36	60	0	0.24 (20%)
S3 film	1.2	0.36	60	0.12 (10%)	0
S4 film	1.2	0.36	60	0.12 (10%)	0.24 (20%)

along with their abbreviation, in detail. As shown in Table 3, the TS of the S1 film was found to be 2.88 ± 0.05 MPa. In the literature, the TS of the control SA film was reported to be 1.23 MPa.[43] In our study, no significant change in elasticity was found. The % elongation at break for all SA films was found to be in the range of 104.46 ± 2.71% to 106.73 ± 0.44%. Thus, the values of % elongation at break were: 105.78 ± 0.11%, 106.09 ± 0.24%, 106.73 ± 0.44%, and 104.46 ± 2.71%, for the S1, S2, S3, and S4 film, respectively. Thus, the incorporation of SiO_2 and propolis extract into SA did not significantly change the % elongation at break of the film.[17] The addition of 0, 5, and 10% citric acid to SA also did not change the % elongation at break. This was attributed to improved intermolecular interactions, which limits the elasticity but improves the rigidity.[44]

3.3. FTIR analysis

As shown in Figure 2(a), all the films had a broad peak at 3244–3285 cm^{-1}. For the S1 film the broad peak at 3244 cm^{-1} corresponds to the hydroxyl (–OH) group's stretching vibrations, while the peak at 2924 cm^{-1} is associated with the stretching vibrations of the –CH bonds. The band at 1594 cm^{-1} signifies the asymmetric vibrations of –COO groups, and the peak at 1407 cm^{-1} is attributed to the symmetric stretching vibrations of –COO groups.[45] The peak at 1301 cm^{-1} is ascribed to CH_2 symmetric twisting vibrations,[46] and glycosidic symmetric vibrations are evident at 1084 cm^{-1}.[47] The peaks at 1024 and 946 cm^{-1} are assigned to the elongation vibrations of the C–O–C groups and out of plane C–H bending vibrations, respectively.[46,47] For the S2 film, changes in wavenumber, peak intensity, and peak broadness were observed. The peak at 3285 cm^{-1} is assigned to the stretching vibrations of hydroxyl and amine groups of the biopolymers (For example, SA, pectin, protein, cellulose, hemicellulose, lignin, etc.). The peak intensity and broadness decreased and the wavenumber slightly increased to 3285 cm^{-1}, indicating changes in hydrogen bonding because of the interaction of the DFP component with SA.[48] The peaks at 1596, 1407, and 1028 cm^{-1} moved to slightly higher wavenumbers and lower peak intensities, indicating an alteration in the asymmetric vibrations of the –COO groups, symmetric stretching vibrations of the –COO group and alteration the in elongation vibrations of the C-O-C group. In the literature similar changes in the FTIR spectra have been reported when a different citrus peel powder was added to SA film.[15] The addition of MgO nanoparticles resulted in only a few changes in the wavelength number of the peaks in the S3 film. The wavelength number for the peaks at 3274, 2928, 1596, 1325, and 1086 cm^{-1} were found to increased, but the peak intensities were lower as compared to the S1 film, which may indicate intermolecular interactions between MgO and SA. For the S4 film, the wavelength numbers increased for the 1092 and 1029 cm^{-1} peaks, while for the 2926 cm^{-1} peak it was slightly decreased as compared to the S3 film. The addition of DFP to the

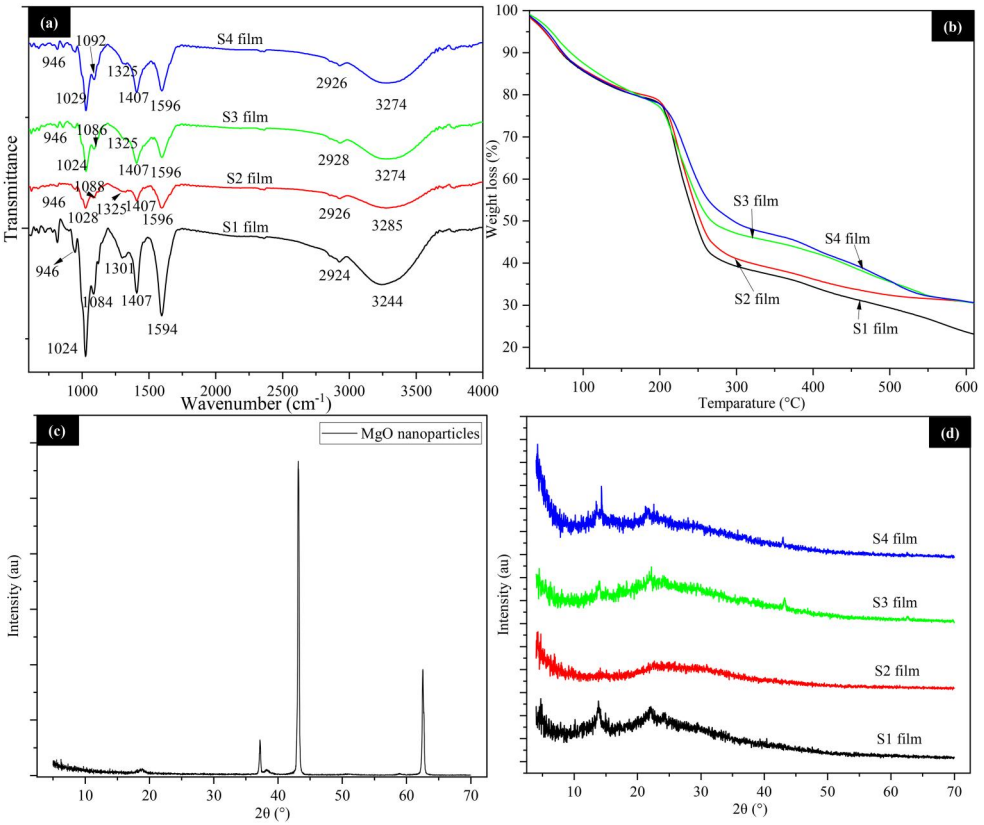


Figure 2. FTIR spectra (a) and TGA curves (b) of the S1, S2, S3, and S4 films, XRD spectra of MgO nanoparticles (c), and S1, S2, S3, S4 films (d).

S3 film resulted in improved intermolecular interactions of the glycosidic symmetric vibrations and C–O–C groups for the S4 film. De Silva et al.[49] reported a similar shift in wavenumber for MgO nanoparticles incorporated in chitosan films and postulated that an increase in interaction might enable the effective transfer of stress from the polymer matrix to the filler particles under tensile loading. This may result in an increase in the tensile strength of the biofilm. For the S2, S3, and S4 films, no new peaks were observed, which indicates the lack of formation of new chemical bonds during the interaction of the DFP, MgO nanoparticles and the SA. Luo et al.[40] reported a lack of chemical bonding when the interaction of MgO nanoparticles, potato starch and tea polyphenols was studied by FTIR analysis. Instead, these researchers reported that MgO nanoparticles bonded with starch by non-covalent interactions, such as hydrophobic interaction, hydrogen bonds, and electrostatic attraction. Similarly, no new peaks were reported during the FTIR analysis of MgO nanoparticles incorporated in polylactic acid biofilms.[22]

3.4. Thermal stability of the SA films

The thermal degradation behavior of the SA films was observed using TGA analysis. The TGA spectra are shown in Figure 2(b). Three distinct degradation stages were

Table 4. Decomposition temperatures of the various SA films at 35% (T_{d35}), 50% (T_{d50}), and 90% (T_{d90}) residual weight, and residual weight (%) at 600 °C.

Sample	T_{d35} (°C)	T_{d50} (°C)	T_{d90} (°C)	Residual weight (%) at 600 °C
S1 film	397	248	77	23.645
S2 film	436	255	80	30.821
S3 film	430	271	89	30.786
S4 film	513	301	79	30.896

observed. The first degradation stage was observed below 150 °C and moisture loss was suggested to be mainly responsible for the mass loss.[50] The second mass loss was observed in the temperature ranges of 150 to 350 °C. This mass loss was attributed to the degradation of the central components of the films, such as SA, glycerol, protein, pectin, cellulose, bioactive compounds, etc.[15] The third stage of degradation, above 350 °C, is related to the gradual oxidative degradation of the carbonaceous residue and lignin.[12] It can be observed from Table 4 that the S4 film was more thermally stable, the thermal decomposition temperatures at 35% and 50% residual weight (%) were higher as compared to the S1, S2, and S3 films. Also, the residual weight increased after the addition of DFP peel powder and MgO nanoparticles. This may be because of the intermolecular interaction between the DFP, MgO, and SA. The data is in line with the tensile strength data (shown in Table 4), in which the S4 film was found to have the highest TS as compared to the other SA films. Luo et al.[40] reported that the proper incorporation of tea polyphenols and MgO nanoparticles can effectively hinder the mobility of starch molecular chains, consequently enhancing the thermal stability of the film. Improvement in the thermal stability of SA films was also observed when various citrus peel powder was added to the SA films.[15] Chitosan/polyvinyl alcohol films also exhibited enhanced thermal stability when incorporating the highest concentration of orange peel powder (1.25% w/w). This improvement was attributed to the formation of robust hydrogen bonding interactions of its powder with the polymer chains, which hindered the evaporation of water molecules and, consequently, prevented chain degradation when compared to the control film.[12]

3.5. XRD analysis

Figure 2(c,d) represent the XRD spectra for the MgO nanoparticles and various SA films within the range of 4° to 69.8° (2θ). The XRD spectra of the MgO nanoparticles showed distinct peaks at angles of 18.34°, 37.2°, 38.2°, 43.16°, and 62.54°. Similar 2θ peak values and XRD spectra for MgO nanoparticles have been reported in the literature.[51] As shown in Table 5, the S1 film exhibited a small peak at 13.85° and an amorphous region with a peak value of 22.11°, consistent with the literature values for pure SA film.[52] The addition of DFP (for S2 film) reduced the CI (%) from 20.41% to 11.81%. Additionally, for the S2 film, the intensity of the 14.12° peak was decreased, while the intensity of the 22.64° peak was increased as compared to the S1 film. These changes may indicate that the DFP disrupted the crystalline conformation of the SA. Schutz et al.[28] also observed a decrease in crystallinity index with the addition of coffee husk powder (5 wt%) into a starch-based biofilm, where it was suggested that the well-dispersed filler particles inhibited the regular molecular packing of the film matrix.[53]

Table 5. Peaks (2θ) and % crystallinity index (CI) of the S1, S2, S3, and S4 films.

Film matrix	Peak (2θ)	CI (%)
S1 film	13.85	20.41
	22.11	
S2 film	14.12	11.81
	22.64	
S3 film	13.92	20.18
	22.2	
	43.23	
	62.64	
S4 film	14.31	28.92
	22.59	
	42.93	
	62.54	

No change in the CI (%) was observed between the S1 and S3 films even though the S3 film had less SA. With the addition of MgO nanoparticles into SA (S3 film), new peaks at 43.23° and 62.64° were observed. The intensity of the amorphous region, with a peak value of 22.2°, was also found to increase. This indicates that the MgO nanoparticles might be promoting the creation of amorphous regions. This increase in amorphous region could offset any increase in crystallinity (caused by new peaks at 43.23° and 62.64°), resulting in no overall change in CI (%). The similar peak angles between the MgO nanoparticles and the S3 film suggest that MgO nanoparticles have a crystallographic structure that is not affected by the SA film matrix. This compatibility may lead to minimal disruption of the existing crystal structure of the SA, keeping the CI (%) unchanged. For the S4 film, similar peak values of 42.93° and 62.54° to those of the MgO nanoparticles were also observed. The intensity at the peak value of 14.31° was found to be higher than those in the S1, S2, and S3 films. The synergistic effect of DFP and MgO nanoparticles are suggested to be responsible for this. Also, the intensity of the amorphous region with a peak value of 22.59°, was found to be similar to that of the S2 film and S3 film. The cumulative effect of MgO nanoparticles and DFP resulted in the S4 film having the highest CI (%), at 28.92%. The enhancement in %CI is in line with the increase in thermal stability of the S4 film reported during the TGA analysis discussion.

3.6. ESEM analysis

Environmental scanning electron microscopy was conducted to understand the microstructure and surface morphology of the SA films; images of the surface of the SA films are shown in Figure 3. The S1 film exhibited a smooth, homogenous and uninterrupted surface without any signs of fracturing. When analyzing the S2 film, it was evident that the surface displayed a more uneven and irregular texture as compared to the S1 film. DFP contains soluble biopolymers, such as protein and pectin, which can contribute to the formation of a closely packed film matrix of SA with a dense internal structure. The higher amount of insoluble components in DFP (such as crude fiber and fat) might distribute randomly in the SA film matrix, which can make the surface rough. In the literature, the rough surface of SA films with lemon peel powder and pomelo peel powder was reported, as lemon and pomelo peel had low

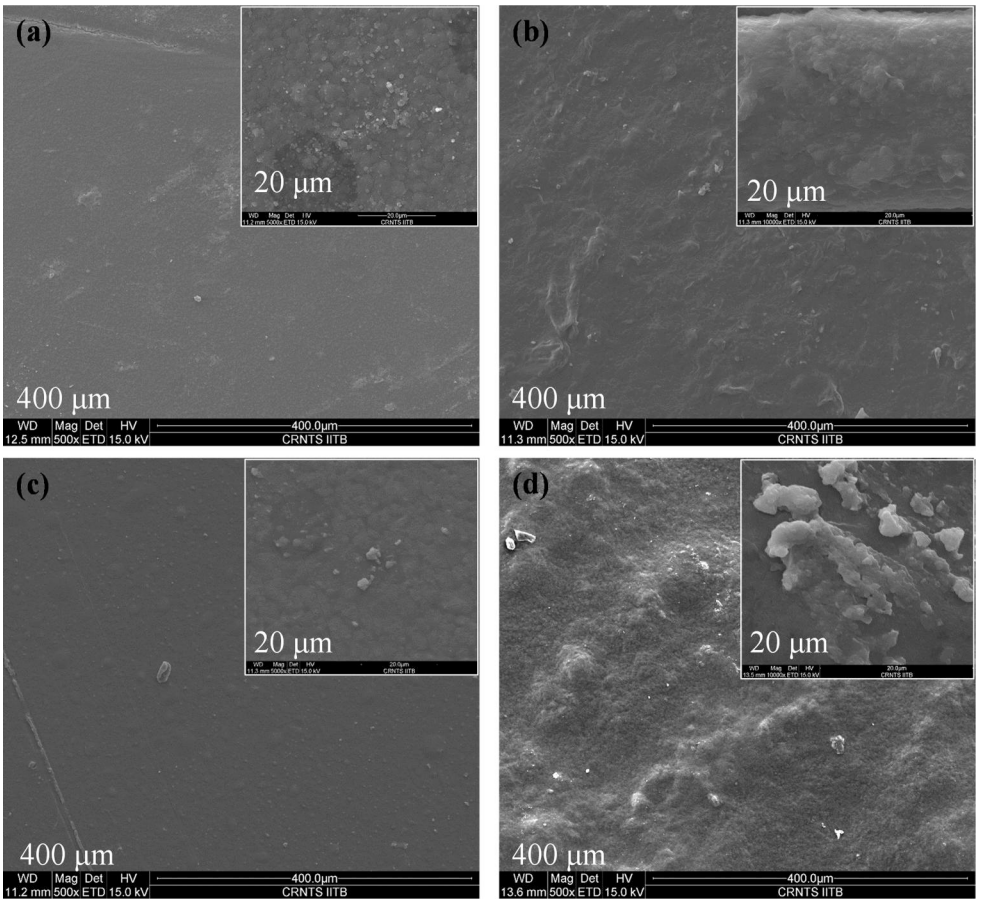


Figure 3. Surface ESEM micrographs of (a) S1 film, (b) S2 film, (c) S3 film and (d) S4 film each with 400 μm and 20 μm scale bars.

pectin and high crude fiber content. Researchers also reported a smooth surface of orange peel powder and mandarin peel powder SA film, as it had a high pectin content.[15] For the S3 film, the film showed a smooth and homogenous appearance at 500x magnification with a white spot in the middle, and a few white spots appeared at 10000x magnification due to the uniform dispersion of the MgO nanoparticles. The few white spots indicate minor aggregation of MgO nanoparticles. Therefore, the MgO nanoparticles were successfully loaded into the SA film. The absence of pores and voids suggests that there was a strong interfacial interaction between the MgO nanoparticles and the SA. Sun et al.[54] observed similar changes in the morphology of silver nanoparticles incorporated SA film. As shown in Figure 3(d), the S4 film had a course and rough surface. The interaction between DFP and MgO nanoparticles may have led to the formation of clusters or uneven distribution, which can result in the rougher surface of the S4 film as compared to the S1, S2, and S3 films. The incorporation of both SiO_2 nanoparticles and propolis extract into SA led to a rougher surface as compared to the control film.[17] The authors attributed the increase in surface roughness to the microscale agglomeration of the SiO_2 nanoparticles.

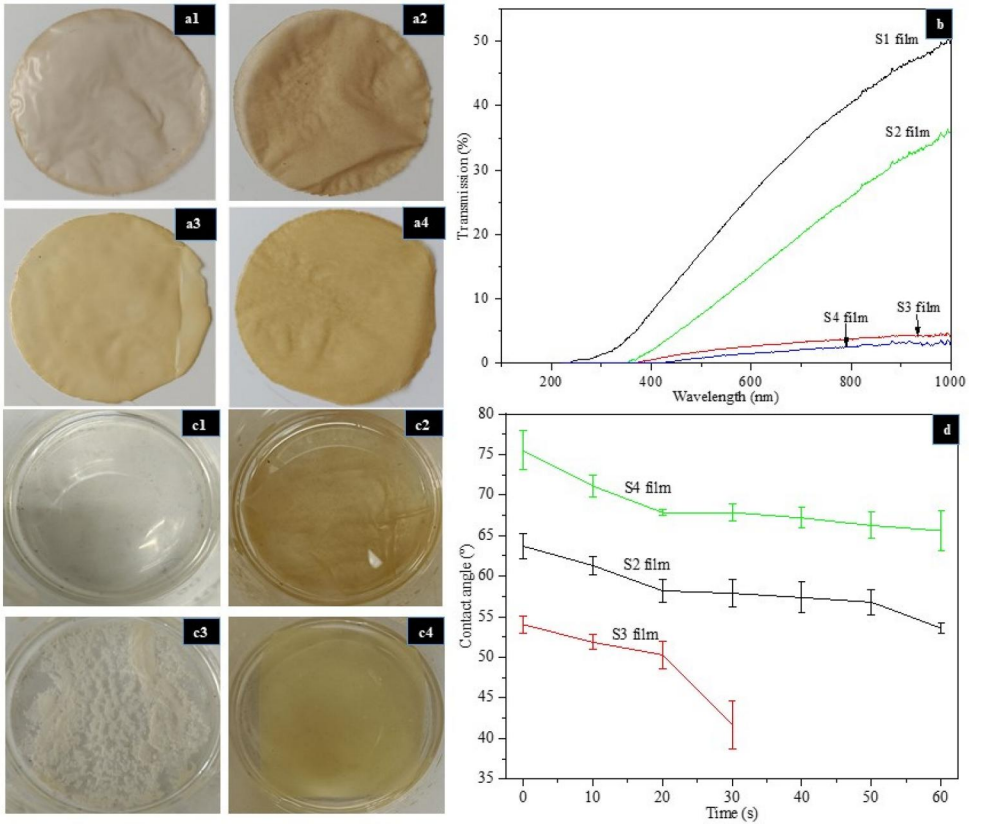


Figure 4. Physical appearance of S1 film (a1), S2 film (a2), S3 film (a3), S4 film (a4), light transmittance (%) of the SA films (b), wet stability study of S1 (c1), S2 (c2), S3 (c3), and S4 (c4) film immersed in water for 20 min, and dynamic contact angle of SA films (d).

3.7. UV-spectra analysis, film opacity and transparency

The physical appearances of S1, S2, S3, and S4 are shown in Figure 4(a1,a2,a3,a4), respectively. The film's ability to act as a barrier against UV and visible spectra is a critical factor in packaging applications. Exposure to UV light can lead to degradation of food products through oxidative processes. Additionally, dairy and food items containing β-carotene can experience degradation due to exposure to visible light.[28] As shown in Figure 4(b), the S1 film exhibited the highest light transmittance in the 200 to 1000 nm wavelength spectra, followed by the S2, S3, and S4 films in order. The S4 film showed zero transmittance in the UV-light region (200–400 nm). Similarly, the S3 and S4 films showed a significant reduction in transmittance in the visible light region (400–700 nm). The significant reduction in transmittance for the S3 and S4 films, we suggest, was because of the presence of the polyphenolic compounds, betalains compounds, lignin, protein, and MgO nanoparticles. The polyphenolic compounds and betalains compounds are known to possess strong UV-radiation absorption capacity because of the presence of C = O and C = C bonds. Also, the light scattering effect may happen due to the presence of the fillers in the polymer matrix.[12] As shown in Table 1, the control film (S1 film) had the highest transparency (measured at 600 nm

wavelength), 26.29 ± 1.31%, while the S4 film was found to have the lowest value of transparency, 1.82 ± 0.05%. The transparencies in the S2 and S3 films were found to be 16.18 ± 0.03% and 3.85 ± 0.05%, respectively, which were lower than that of the S1 film. The opacity of the S1 film was 13.58 ± 0.54, and it increased up to 20.14 ± 0.61 AU nm/mm for the S4 film. Hou et al.[55] also reported a similar trend of transparency and opacity when varying concentrations of SiO_2nanoparticles (2.5, 5, 7.5, and 10 wt%) were added to agar/SA film. The higher opacity indicates good light barrier properties. The opacity and transparency were found to be inversely proportional to each other. The optical characteristics of thin films are intricately linked to the functional groups present within the filler particles, exerting a pronounced influence on the interfacial dynamics governing the interactions between the polymer chains and the filler surfaces.[56,57] The findings are consistent with previous research in which starch-based films were reinforced with small-sized corn and wheat hulls. The researchers observed that the small-sized particles dispersed more uniformly within the starch matrix, improved the UV-blocking properties, and decreased the transparency of the film.[58] In the current study a similar effect could have occurred, where the DFP and MgO nanoparticles were effectively dispersed in the SA matrix, leading to the observed effect on the biofilm's light properties. Terzioglu et al.[12] observed a decrease in the transparency of chitosan/PVA with the incorporation of orange peel powder. The decrease in transparency was mainly attributed to the light scattering effect due to the addition of the peel powder as a filler into the polymer matrix. With increasing carrot fibers concentration (1, 3, and 5%) in chitosan films, the transparency decreased from 77.80% to 41.43%, while the opacity increased from 710.1 to 1662.5 AU nm/mm.[59] Therefore, the synergistic effect of the addition of DFP and MgO nanoparticles could significantly enhance the light barrier properties of the SA films.

3.8. Dynamic contact angle and wet stability of the SA films

SA has very high hydrophilicity and, because of this, its applications remain limited. Therefore, our research focused on improving the hydrophobicity. The contact angle of the S1 film could not be measured because of high swelling tendency and hydrophilicity. As shown in Figure 4(d), the addition of DFP increased the contact angle of the SA films from 0 to 63.7 ± 1.5°. The MgO nanoparticles addition (S3 film) also improved the contact angle, to 54.05 ± 1.06°. For the S4 film, the synergistic effect of DFP and MgO nanoparticles increased the contact angle to 75.5 ± 2.4° as compared to the S1 film. Due to the swelling of the S3 film, the contact angle could not be measured after 40 s time. The swelling problem did not occur in the S2 and S4 film during measurement of the contact angle. As shown in Figure 4(c), the stability of the SA films in water was evaluated, in which the morphology of the film was observed after immersing the film in the water for 20 min at room temperature. The S1 film was found to be completely dissolved in the water. The S3 film disintegrated into small pieces and was barely visible. The S2 film was completely integrated, and retained its shape. The S2 film was thus less affected by water as compared to the S1 and S3 films. However, the swelling of the S2 film was observed. Similarly, the S4 film also showed no disintegration and remained in its shape. Therefore, we conclude that the DFP can significantly

improve the wet stability of the SA film. The presence of lignocellulose[60] and bioactive compounds[61] can enhance the water resistance properties of the polymer matrix, resulting in excellent wet stability properties. Similar to our observation, Liu et al.[62] reported the complete dissolution of their SA control film, and that the addition of graphene oxide and ammonia-functionalized graphene oxide greatly improved the wet stability of their SA composite films.

3.9. Thickness, moisture content (%) and water vapor permeability

As shown in Table 1, the thickness of the S1 film was 0.0540 ± 0.0026 mm. Because of the peel compounds, such as crude fiber, protein, and bioactive compounds, the thickness of the S2 film increased to 0.0686 ± 0.0015 mm. For the S3 film, the thickness was found to be 0.1033 ± 0.01387 mm. Because of the synergistic effect of the MgO nanoparticles and peel powder compounds, it was observed that the S4 film had the highest value of thickness, 0.1066 ± 0.0083 mm. The thickness values of the S3 film and S4 film were in a similar range. As compared to the S1 film, significant increases in the thickness values were observed for the S3 film, and S4 film. The MC for the control film (S1 film) was $20.93 \pm 0.77\%$. With the addition of DFP (S2 film), the MC was found to decrease slightly, to $19.35 \pm 0.77\%$. The hydrophobic nature of lipid has been reported to decrease the moisture content of a film containing it.[58] DFP has a crude fat content of 1.66%, which contributed to decreasing the MC. The MC of SA film incorporating lemon peel powder (LP) and pomelo peel powder (PP) was reported to be 25.25% and 22.66%, respectively.[15] The authors observed the lower value of lipid content in LP and PP as the primary reason for the higher values of MC. Moreover, the SA films with orange peel powder containing high lipid content led to a low MC of 18.16%.[15] The addition of MgO nanoparticles (S3 film) resulted in an increase in the value of MC to $24.14 \pm 0.81\%$; this may be because of the known hydrophilic nature of the MgO nanoparticles. For the S4 film, the synergistic effect of MgO nanoparticles and DFP led to a modest decrease in MC, to $21.00 \pm 0.19\%$. The hydrophobic properties of crude fat and lignin present in DFP counteracted the hydrophilic property of the MgO nanoparticles. This interaction created a balance in overall hydrophilicity of the S4 film. Therefore, this interaction led to a modest decrease in MC of the S4 film as compared to the S3 film. However, the value of MC for the S4 film was slightly more than that of the control film, and the hydrophilic nature of the MgO nanoparticles could be responsible for this.

From Table 1 the WVP of the SA films can be observed. The addition of DFP into the SA film resulted in a decrease in the WVP, from 3.55 to 3.28×10^{-10} $g \cdot m^{-1} \cdot s^{-1} \cdot Pa^{-1}$. The filler particles can increase the tortuosity of the water molecule pathway.[63] However, for the S3 film, there was an increase in WVP, to 3.63×10^{-10} $g \cdot m^{-1} \cdot s^{-1} \cdot Pa^{-1}$. We attribute this increase to the aggregation of the MgO nanoparticles, which decreased the tortuosity and may have caused the separation of the polymer chains, thereby creating a wider pathway for water molecules to pass through. Furthermore, the synergistic effect of the MgO nanoparticles and DFP decreased the WVP to 3.45×10^{-10} $g \cdot m^{-1} \cdot s^{-1} \cdot Pa^{-1}$ for S4 film. For the SA film incorporated with Gallnut extract, Aloui et al. (2021)[64] observed that crosslinking between phenolic compounds and the

polymer chains decreases the free volume in a polymer matrix, therefore creating a denser polymer matrix. This can decrease the WVP value of the S4 film. However, due to aggregation of the MgO nanoparticles, the WVP value of S4 film was higher than that of the S2 film. We note these values are lower than the reported literature values for SA films incorporating various materials: sulfur nanoparticles ($8.7–11.8 \times 10^{-10}$ $g{\cdot}m^{-1}{\cdot}s^{-1}Pa^{-1}$),[65] oregano essential oil ($27–31 \times 10^{-10}$ $g{\cdot}m^{-1}{\cdot}s^{-1}Pa^{-1}$),[66] and SiO_2 nanoparticles ($3.9–5.0 \times 10^{-10}$ $g{\cdot}m^{-1}{\cdot}s^{-1}Pa^{-1}$).[52]

3.10. Antioxidant activity

As shown in Table 1 the values of the ABTS radical scavenging activity (%) of the SA films increased significantly with the addition of MgO nanoparticles and DFP. For the S1 film, the ABTS radical scavenging activity (%) was $22.53 \pm 1.47\%$. In the literature the values of ABTS radical scavenging activity of SA/maltodextrin and pectin/SA/xanthan gum film were reported to be 25% and 35.31%, respectively. We mainly attribute the presence of phenolic,[67] and betalain pigment[68] compounds in the DFP for the increase in ABTS radical scavenging activity (%) of the S2 film to $39.12 \pm 1.95\%$. The polyphenolic compounds exhibit antioxidant activity due to the phenolic ring's ability to delocalize the unpaired electron and to the donation of hydrogen atoms by the hydroxyl groups.[67] The addition of *Azollapinnata* leaves extract also increased the antioxidant activity of SA/maltodextrin films. The authors attributed the effect to scavenging of the free radicals caused by the presence of polyphenols inside the films.[9] For the S3 film, the value of the ABTS radical scavenging activity (%) increased to $73.48 \pm 0.52\%$. Bhakya et al.[69] reported that their improvement in antioxidant activity by the addition of nanoparticles was related to the functional groups of molecules attached to the surfaces of the nanoparticles. MgO nanoparticles can facilitate the electron transfer and neutralize the free radicals. Moreover, the uniform dispersion and higher surface area of MgO nanoparticles can influence the antioxidant properties. Hosseini et al.[70] reported a similar trend of increase in values of antioxidant activity of starch/albumin film with the addition of MgO nanoparticles. The combined effect of DFP and MgO nanoparticles (S4 film) resulted in a significant increase in the ABTS radical scavenging activity (%), to $96.16 \pm 1.57\%$. Qin et al.[71] reported that the chitosan/purple corn extract/Ag nanoparticle film had higher antioxidant activity as compared to the chitosan/Ag nanoparticle and chitosan/purple corn extract films. A similar increase in ABTS radical scavenging activity (up to 87.93%) was reported because of the synergistic effect of ZnO nanoparticles and bamboo leaves extract in a chitosan film.[72]

3.11. Overall migration (OM) study of SA films

The migration refers to the movement of chemical compounds from the packaging biofilm material into the simulating fluid. The OM measures the total amount of non-volatile compounds that can migrate to food substances. The OM study was conducted in accordance with EN-1186-1 guidelines to assess the safety of the SA films. The current study used the simulants D1 (50% ethanol) and D2 (i-octane) to simulate contact with

alcoholic beverages and fatty food. As shown in Table 2, all of the SA films showed much higher OM values than the maximum permissible limit of 60 mg/kg of simulant for 50% ethanol simulant exposure. The SA has a strong affinity toward water due to its hydrophilic nature. Ethanol, being a polar solvent, might also contribute toward the migration of polar substances from the film to the simulant. This may result in high values of OM for 50% ethanol simulant fluid. The OM values of red grape extract incorporated soy protein isolate film were reported to be in the range of 1622 ± 220 to 2387 ± 590 mg/kg of distilled water simulant for 4, 20, and 40 °C temperature. The higher values of OM, compared to the maximum permissible limit, were attributed to the migrant-matrix affinity and the migrant solubility toward simulants.[73] The liberation of active compounds from the polymer network takes place by diffusion of solvent (such as 50% ethanol and isooctane) into the film matrix from the outer solution, followed by the weakening of the polymer network. These changes lead to active compounds' diffusion into the outer solvent until the thermodynamic equilibrium is achieved. Therefore, the OM values are intrinsically reliant upon such factors as matrix solubility, the progressive infiltration of solvent into the film matrix and the concurrent diffusion of active compounds into the surrounding solvent medium.[74] All the SA films complied with the OM limit of 60 mg/kg of simulant for the isooctane simulant, in accordance with the EN-1186. Therefore, all SA films are safe to use in fatty food environments. The observed phenomenon in the case of isooctane can be attributed to the limited extent of isooctane diffusion into the hydrophilic polymer matrix, alongside the preservation of the polymer's closed structural integrity.[74] Our extensive literature research yielded no reports on OM values for SA films. However, similar results for soy protein isolate film were reported for isooctane simulants.[73] For chitosan nanofibers incorporated poly(butylene adipate-co-terephthalate) films, OM values less than 30 mg/kg of simulant for isooctane simulant were also reported.[31]

3.12. Biodegradation of the SA composite films

As shown in Figure 5, a biodegradation study in organic compost was conducted to assess the biodegradability of our SA films. The results are presented in Table 6, which illustrates the percentage change of mass loss observed at regular time intervals. The S1 film exhibited significant degradation with a mass loss of 65.70% observed on day 5. It was completely degraded by day 10. The S2 film displayed a remarkable 82.09% mass loss on day 5, indicating rapid degradation during the initial period. However, it continued to degrade very slowly for the next 20 days, maintaining its physical appearance. In contrast to the S1 and S2 films, the S3 film showed a slower rate of degradation. On day 5, only 8.36% mass loss was observed, increasing to 26.79% on day 10 and 67.06% on day 15. Complete degradation in soil was achieved by day 20. Furthermore, on day 15, the S3 film started to crumble into small pieces. The S4 film displayed a mass loss of 25.51% on day 5, more mass loss than the S3 film (8.36%) on day 5.

Furthermore, the mass loss of the S4 film increased to 39.88% and 51.02% for day 10 and day 15. The S4 film achieved complete degradation by 20 days. In the literature, complete degradation of brown rice starch incorporated chitosan film in 20 days was reported.[32] Similarly, SA-starch-montmorillonite films were reported to be completely

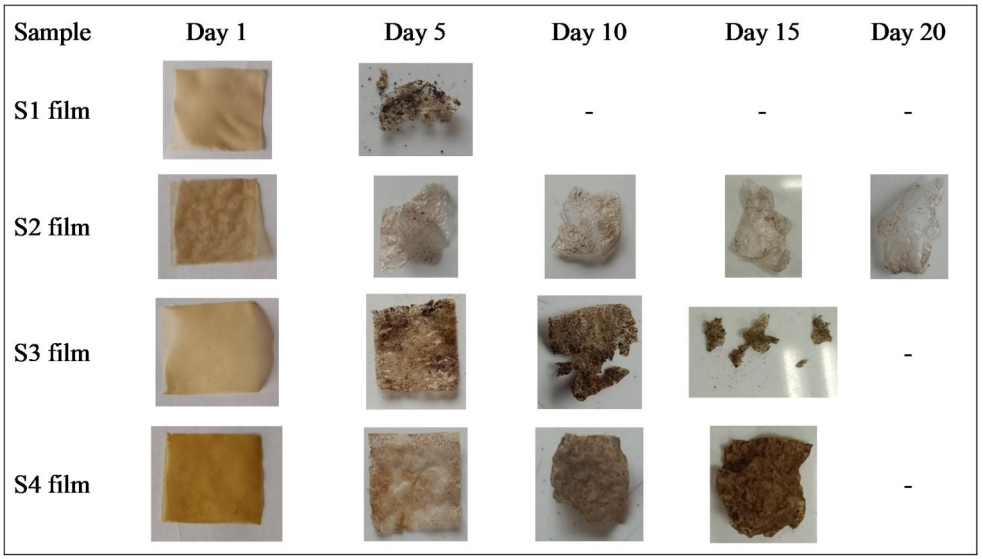


Figure 5. Physical appearance of films on day 1, 5, 10, 15, and 20.

Table 6. Mass loss of the various SA films over a period of 20 days at room temperature.

Sample	Mass loss (%) Day 1	Mass loss (%) Day 5	Mass loss (%) Day 10	Mass loss (%) Day 15	Mass loss (%) Day 20
S1 film	0	65.70	100	100	100
S2 film	0	82.09	84.26	84.99	84.99
S3 film	0	8.36	26.79	67.06	100
S4 film	0	25.51	39.88	51.02	100

degraded in 22 days in the soil during a biodegradation study.[75] It is crucial to emphasize that all samples initiated their degradation prior to reaching 180 days, which is the timeframe required for at least 90% of the plastic material to disintegrate in order to qualify as biodegradable (ASTM D640090).

4. Conclusions

Bioactive films based on SA, DFP, and MgO nanoparticles were successfully fabricated. ESEM images showed that the addition of the MgO nanoparticles maintained the smooth surface of the film. However, the crude fiber present in the DFP led to an increase in the surface roughness. FTIR analysis confirmed the molecular interaction between DFP, MgO nanoparticles, and SA. The S4 film had the highest thermal stability, tensile strength, thickness, and film opacity. ABTS radical scavenging activity increased significantly for the S4 film, to 96.16%, as compared to the S1 film (22.53%). The film with DFP had the lowest WVP, 3.28×10^{-10} $g^{-1}\,Pa^{-1}\,S^{-1}$. In accordance with EN-1186-1 guidelines, the OM values for the SA films were less than 60 mg/kg of simulant for a fatty food environment. However, the OM values surpassed this threshold when the films were exposed to 50% ethanol simulant, rendering the SA films unsuitable for application in alcoholic beverages. Additionally, the films exhibited a noteworthy degree of biodegradability when subjected to soil conditions.

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Authors' contributions

Rushikesh A. Prajapati: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – Review & Editing; Girirajsinh C. Jadeja: Corresponding author, Supervision, Conceptualization, Resources, Review and Editing.

Disclosure statement

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